

Senders and Coroutines

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Abstract

`std::execution` serves its domain well. Different asynchronous domains have different costs, and a single model cannot minimize all of them simultaneously. This paper identifies four structural gaps where the sender model meets coroutines: three at the boundary - error reporting, error returns, and frame allocator propagation - and one inside the composition mechanism - the symmetric transfer gap documented in P2583R0^[63]. Each gap is the cost of a property the sender model requires for compile-time analysis (P2300R10^[1], P4014R0^[26]). They are not design defects, they are tradeoffs. Mandating that standard networking be built on the sender model would force coroutine I/O users to pay these costs. A coroutine-native I/O research report (P4003R0^[25]) made the gaps visible by showing that partial results, error returns, cancellation, and frame allocator propagation emerge naturally when I/O is designed for coroutines. The findings hold regardless of that report's specific design. This paper recommends: ship `std::execution` for C++26, defer `task` (P3552R3^[13], "Add a Coroutine Task Type") to C++29, and explore coroutine-native I/O designs alongside sender-based designs.

Revision History

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- Initial version.
-

1. Disclosure

The authors developed P4003R0^[25] ("Coroutines for I/O"). A coroutine-only design cannot express compile-time work graphs, does not support heterogeneous dispatch, and assumes a cooperative runtime. Those are real costs.

The structural gaps documented in Sections 3-5 exist at the boundary where `std::execution` meets coroutines. Section 6 summarizes a fourth gap - inside the composition mechanism - documented in full by P2583R0^[63]. All four gaps hold regardless of whether P4003 is the right design or any design at all. Code examples in Section 2 are drawn from P4003 - not to propose it as the only way forward.

2. Coroutine-Native I/O

Eric Niebler wrote in "Structured Concurrency"^[33] (2020): *"I think that 90% of all async code in the future should be coroutines simply for maintainability."*

If coroutines are the primary tool, what does I/O look like when designed for them? P4003R0^[25] ("Coroutines for I/O") is a research report drawn from a working implementation compiled on three toolchains, with benchmarks and unit tests. It produced a minimal protocol - two concepts, a type-erased executor, and thread-local frame allocator propagation - and identified four properties of the coroutine-native approach.

2.1 Partial Results

A composed I/O completion returns an error code and a result together:

```
auto [ec, n] = co_await read(sock, buf);
```

As Peter Dimov observes, `(error_code, size_t)` is semantically equivalent to two consecutive calls: the first returning `({}, n)`, the second returning `(ec, 0)`. The tuple carries the full history of the operation. A `read` that transferred 47 bytes before EOF gives the caller 47 bytes and `eof`. Error and byte count travel together. The coroutine decides what to do with both.

2.2 Error Returns

`co_return` handles success and failure uniformly:

```
route_task https_redirect(route_params& rp)
{
    std::string url = "https://";
    url += rp.req.at(field::host) + rp.url.encoded_path();
    rp.res.set(field::location, url);
    auto [ec] = co_await rp.status(status::found).send("redirect");
    if (ec)
        co_return route_error(ec);
    co_return route_done;
}
```

cppcoro, folly::coro, Boost.Cobalt, Boost.Asio, libcoro, and asyncpp use the same convention.

2.3 Cancellation

On Windows, `CancelIoEx` completes the pending operation with `ERROR_OPERATION_ABORTED`. On POSIX, cancellation produces `ECANCELED`. The error code arrives through the same field as every other error, accompanied by the same byte count. Code that checks `ec` handles cancellation without additional logic.

2.4 Frame Allocator Propagation

Coroutine frame allocation overhead is measurable at high request rates. A recycling frame allocator eliminates it (Section 5 benchmarks a 3.1x speedup on MSVC). The frame allocator is set once at the launch site:

```
run_async( &pool )( do_connection(sock) );
```

Thread-local propagation delivers the frame allocator to every coroutine in the chain without polluting signatures. P4003R0^[25] Section 5 provides the complete mechanism: the `operator new` hook, the resume-time restore, and the timing guarantee from C++17 evaluation order. Coroutine signatures do not mention the frame allocator:

```

route_task serve_api(route_params& rp)
{
    auto result = co_await db.query("SELECT ..."); // uses frame allocator here
    auto json = serialize(result);
    auto [ec] = co_await rp.send(json);           // and here
    if (ec) co_return route_error(ec);
    co_return route_done;
}

```

The frame allocator is set once at the launch site and reaches every coroutine automatically. We would welcome the opportunity to work with the P2300R10^[1] and P3552R3^[13] authors to explore whether P4003R0^[25]'s thread-local propagation mechanism could deliver automatic frame allocator propagation to `std::execution::task`.

2.5 Two Models of Computation

The Sender Sub-Language expresses the same logic differently:

```

auto sndr = just(std::move(socket))
| let_value([](tcp_socket& s) {
    return async_read(s, buf)
    | then([](auto data) {
        return parse(data); // VALUE -> success path
    });
})
| upon_error([](auto e) { // ERROR -> error path
    log(e);
})
| upon_stopped([] { // STOPPED -> cancellation path
    log("cancelled");
});
auto op = connect(std::move(sndr), rcvr);
start(op);

```

Niebler asked the question himself^[33]: ***“Why would anybody write that when we have coroutines? You would certainly need a good reason.”***

This is **continuation-passing style**^[39] expressed as composable value types. P4014R0^[26] (“The Sender Sub-Language”) provides the full treatment. Coroutines inhabit direct-style C++ - values return to callers, errors propagate through the call stack, resources scope to lexical lifetimes. This paper calls that **regular C++**.

SG4 polled at Kona (November 2023) on P2762R2^[4] (“Sender/Receiver Interface For Networking”):

“Networking should support only a sender/receiver model for asynchronous operations; the Networking TS’s executor model should be removed”

SF	F	N	A	SA
5	5	1	0	1

Consensus.

The poll presented two alternatives: the Networking TS’s executor model and the sender model. A coroutine-native approach was not among the choices. This paper introduces a third option.

This paper recommends: ship `std::execution` for C++26, defer `task` to C++29, and explore coroutine-native I/O designs alongside sender-based designs. The following four sections explain why.

3. Where Do Errors Go?

How should an asynchronous operation report its outcome? The sender model has an elegant answer.

3.1 Senders Use Channels

The committee adopted three Sender completion channels: `set_value`, `set_error`, and `set_stopped`. These channels are the sender model’s type system. Type-level routing is a requirement of CPS reification (Appendix A.1). From P2300R10 Section 1.3.1^[1]:

“A sender describes asynchronous work and sends a signal (value, error, or stopped) to some recipient(s) when that work completes.”

The three channels enable compile-time routing: `upon_error` attaches to the error path at the type level, `let_value` chains successes, and algorithms like `when_all`^[50] cancel siblings when a child completes through the error or stopped channel - all without runtime inspection of the payload.

Each channel has a fixed signature shape:

- `set_value(receiver, Ts...)` carries the success payload.
- `set_error(receiver, E)` carries a single, strongly-typed error.
- `set_stopped(receiver)` is a stateless signal.

3.2 I/O works with tuples

A composed read accumulates bytes across multiple receives. In a coroutine, this is expressed naturally as a tuple (here, `pair`):

```

task<std::pair<std::error_code, std::string>>
read(tcp::socket& sock)
{
    std::string result;
    char buf[1024];
    for (;;)
    {
        auto [ec, n] = co_await sock.read_some(buf);    // also a tuple
        result.append(buf, n);                          // n bytes arrived regardless of ec
        if (ec)
            co_return {ec, std::move(result)};
    }
}

```

The caller always receives both the error and the accumulated data. Composed I/O operations produce tuples because partial progress is the norm. A `read` that fills 47 bytes before the connection resets returns both the error code and the partial data, with no loss. [Boost.Asio](#)^[40] codified this as `void(error_code, size_t)` twenty-five years ago. [P2762R2](#)^[4] (“Sender/Receiver Interface for Networking”) preserves it.

I/O completions are **complicated success**: EOF with complete data, connection reset with accumulated progress, a missing TLS shutdown; these are not failures, they are outcomes the caller must inspect. Even cancellation is an error code (`ERROR_OPERATION_ABORTED` on Windows, `ECANCELED` on POSIX): code that handles errors already handles cancellation for free.

The sender model offers three **simple channels**: one value, one error, one stop signal.

Complicated success does not fit in a simple channel.

3.3 Published Sender Code Loses Data

Every published sender implementation of I/O that the authors could find loses partial results on the error path.

[P2300R10 Section 1.3.3](#)^[1] presents a composed read in its motivating examples. The operation reads a length-prefixed byte array: first the size, then the data.

```

using namespace std::execution;

sender_of<std::size_t> auto async_read(
    sender_of<std::span<std::byte>> auto buffer,
    auto handle);

struct dynamic_buffer {
    std::unique_ptr<std::byte[]> data;
    std::size_t size;
};

sender_of<dynamic_buffer> auto async_read_array(auto handle) {
    return just(dynamic_buffer{})
        | let_value([handle] (dynamic_buffer& buf) {
            // P2300R10 writes as_writable_bytes.
            // The standard library function is std::as_writable_bytes
            return just(std::as_writable_bytes(std::span(&buf.size, 1)))
                | async_read(handle)
                | then(
                    [&buf] (std::size_t bytes_read) {           // from set_value
                        assert(bytes_read == sizeof(buf.size));
                        buf.data = std::make_unique<std::byte[]>(buf.size);
                        return std::span(buf.data.get(), buf.size);
                    })
                | async_read(handle)
                | then(
                    [&buf] (std::size_t bytes_read) {           // from set_value
                        assert(bytes_read == buf.size);
                        return std::move(buf);
                    })
                );
        });
}

```

No coroutines appear in this example. The data loss occurs in the sender pipeline itself.

`async_read` completes through `set_value` with the byte count, or through `set_error` with the error code. Consider the second `async_read` failing partway through. Some bytes transfer. The connection resets. `set_error` fires. The byte count - how far the second read got - has no channel. `then` handles only `set_value`. The error propagates past it, exits `let_value`, and the `dynamic_buffer` with its valid `size`, allocated `data`, and partial contents is destroyed.

This is P2300R10 Section 1.3.3^[1]'s motivating example of sender composition.

The `stdexec`^[52] repository contains a production adaptor for this problem. Robert Leahy's `use_sender`^[60] (February 2026) bridges Asio's `(error_code, Args&&...)` completions into the three-channel model. Leahy is the author of P3373R2^[12] and P3950R0^[23], a contributor to the `stdexec` reference implementation, and an experienced sender/receiver practitioner. The adaptor receives the error code and byte count together from Asio. In its `set_error` path, the error code is transmitted and the remaining arguments - including the byte count - are discarded.

To our knowledge, no published sender code - P2300R10^[1], `stdexec`^[52], Leahy's adaptor - discriminates the channel based on bytes transferred.

3.4 Everything Through `set_value`

The natural response to the data loss in Section 3.3 is to route every I/O completion through `set_value(rcvr, ec, n)`. No data is lost. Both values travel together. The coroutine receives them as a pair:

```
do_read(sock, buf)
| then([](std::error_code ec, std::size_t n) {
    // ec and n both available - no loss
})
| upon_error([](auto e) {
    // never called - I/O errors travel in the pair
})
| upon_stopped([] {
    // never called - cancellation is in the pair
});
```

The consequence is that `let_error`, `upon_error`, `let_stopped`, `upon_stopped`, `stopped_as_optional`, and `stopped_as_error` become inaccessible to I/O and thus cannot participate in the composition promised by the Sender Sub-Language. These are framework primitives; a retry combinator built on `let_error` silently does nothing for I/O senders.

A Note on `when_all` and `when_any`

The combinators `when_all` and `when_any` present a different challenge: they must decide whether a partial result constitutes success or failure, and that decision is domain-specific. I/O is the domain where partial success is pervasive, and no combinator can make that classification without domain knowledge. This paper does not claim those combinators should work differently for I/O. The cost documented here is narrower: six standard algorithms that are unconditionally unreachable.

The generic composition that justifies having three channels cannot serve I/O under this mapping.

3.5 The Classification Problem

Andrzej Krzemiński observes that I/O completions flow through a pipeline with a classification step between the I/O layer and the program logic:

Level:	I/O	Composed I/O	Classification	Program Logic
success:	[read] ---->	[process] ----->	[classify] ----->	[handle success]
error:	(unused)	(unused)	(unused)	\---> [handle error]
cancel:	(unused)	(unused)	(unused)	\--> [handle cancel]

At the I/O level, there is only success. The error and cancellation channels become relevant only after the application classifies the I/O status - and that classification requires context the I/O layer does not have. For example, `ECONNRESET` means “fatal, abort the transaction” in an HTTP request handler, but “done, expected closure” in a long-polling connection that the server intentionally drops. The same error code, classified differently depending on the protocol state the coroutine holds.

`std::execution` does not provide a standard classification adaptor that maps I/O status tuples into the three channels.

Sections 3.6 through 3.8 examine what happens when the mapping is attempted.

3.6 Dimov's Mapping

Peter Dimov proposes a refined convention that avoids data loss on partial success by discriminating the channel based on the byte count and error code category:

Completion	Channel
<code>(eof, n)</code> for any <code>n</code>	<code>set_value(eof, n)</code>
<code>(canceled, n)</code> for any <code>n</code>	<code>set_stopped()</code>
<code>(ec, 0)</code> where <code>ec</code> is not eof	<code>set_error(ec)</code>
<code>(ec, n)</code> where <code>n > 0</code>	<code>set_value(ec, n)</code>

This is the only mapping the authors are aware of that preserves partial results on the error path. Dimov argues that cancellation means the result is no longer needed, so discarding bytes on the stopped path is correct by design. This paper accepts that characterization. EOF is routed through `set_value` regardless of `n`, consistent with I/O's treatment of EOF as a normal outcome rather than a failure.

The mapping requires semantic knowledge of specific error conditions: EOF is distinguished from cancellation, which is distinguished from all other errors. It is not a generic transformation, it is domain-specific I/O logic embedded in the channel routing. Dimov characterized the convention as “ad hoc.”

3.7 Routine Errors Become Exceptions

Under Dimov's mapping, any I/O error with zero bytes transferred goes through `set_error`. P3552R3^[13]'s `task` delivers `set_error` to the coroutine as a thrown exception. Three design decisions chain together:

1. SG4 mandate (Kona 2023): Networking operations must be senders. `s.async_read_some(buf)` returns a sender.
2. Dimov's mapping: `(ec, 0)` where `ec` is not EOF completes through `set_error(ec)`.
3. P3552R3's `task`: When a coroutine `co_await` s a sender, `await_transform` calls `as_awaitable`, which creates a receiver. The receiver's `set_error` handler stores the error as `exception_ptr`. On resume, `await_resume` rethrows. There is no other delivery mechanism - `co_await` can only return a value or throw.

Common transient I/O errors that routinely arrive with zero bytes on a given call:

- `ECONNRESET` - client disconnected
- `ECONNABORTED` - connection aborted
- `ETIMEDOUT` - connection timed out

- `EPIPE` - broken pipe on write
- `ECONNREFUSED` - connection refused
- `ENETUNREACH` - network unreachable

A server handling thousands of connections sees `ECONNRESET` constantly - clients disconnect, networks flap, load balancers probe, mobile users lose signal. Under this model, every routine disconnection that transfers zero bytes on the final call becomes an exception with stack unwinding.

This contradicts twenty-five years of established I/O practice. Kohlhoff designed `Boost.Asio`^[40] with error codes precisely because I/O errors are not exceptional - they are the normal texture of network programming. The C++ I/O ecosystem is built on `if (ec)`. The combination of the SG4 mandate, Dimov's mapping, and P3552R3's `task` reverses this for every error that arrives with zero bytes transferred.

In a coroutine-native design, these same errors arrive as error codes:

```
auto [ec, n] = co_await s.read_some(buf); // ec == ECONNRESET, n == 0
if (ec) co_return {ec, std::move(body)}; // no exception, no stack unwinding
```

3.8 Coroutines Always Pay

Three solutions have been presented. Each improves on the last. None is free:

Solution	Cost
<code>set_error</code> on error	Partial results destroyed
<code>set_value</code> always	Reduced composition surface
Dimov's mapping	<code>co_yield with_error</code> required Routine errors become exceptions (Section 3.7) Same error bifurcates on <code>n</code>

Under Dimov's mapping - the best known convention - consider the same `read_body` from Section 3.2, written with `std::execution::task`. Preserving accumulated data on the `n==0` error path requires try/catch in a normal I/O loop for errors that are not exceptional:

```

std::execution::task<std::string>
read_body( tcp_socket& sock )
{
    char buf[1024];
    std::string body;
    while (true) {
        try {
            auto [ec, n] = co_await sock.async_read_some(buf);
            body.append( buf, n );
            if (ec == eof)
                co_return body;
            if (ec)
                /* signal error to sender pipeline (Section 4) */;
        } catch (...) {
            co_return body;
        }
    }
}

```

The coroutine-native invocation:

```

auto [ec, body] = co_await read_body(sock); // no exceptions, no data loss

```

In a sender pipeline, these costs do not exist. The channels route completions at the type level, and the pipeline pays nothing. At the coroutine boundary, the same channels impose `co_yield with_error`, exception handling for routine I/O errors, and bifurcation of the same error depending on bytes transferred.

For senders, no cost. For coroutines, everything.

3.9 The Paper Not Polled

Chris Kohlhoff identified this tension in 2021 in [P2430R0](#)^[3] (“Partial success scenarios with P2300”):

“Due to the limitations of the `set_error` channel (which has a single ‘error’ argument) and `set_done` channel (which takes no arguments), partial results must be communicated down the `set_value` channel.”

Kohlhoff presented these observations during the P2300 review and published the slides as P2430R0. It was not a standalone proposal, and there were no polls to find. But the observation was right. Five years and ten revisions later, the tension Kohlhoff identified remains unresolved. Appendix A.3 traces why.

3.10 LEWG Background

A companion document^[61] captures the design space and trade-offs around how asynchronous I/O completion results interact with the three-channel model of `std::execution` (P2300R10^[1]). It is intended for LEWG consumption as background for decisions concerning `std::execution::task`, the SG4 networking mandate, and the relationship between senders and coroutines.

4. Where Is the `co_return` ?

The three-channel completion model requires coroutines to deliver errors through a path that C++ coroutines do not provide. The workaround requires a language change.

4.1 The Mechanism

In a sender's operation state, signaling an error is direct: `set_error(std::move(rcvr), ec)`.

In a coroutine, `co_return expr` calls `promise.return_value(expr)`, which P3552R3^[13] routes to `set_value`. There is no way to make `co_return` call `set_error`.

A coroutine promise can define `return_void` or `return_value`, not both ([dcl.fct.def.coroutine]^[50]). `task<void>` needs `return_void`. The `return_void` / `return_value` mutual exclusion blocks any `return_value` overload that could route to `set_error`.

`co_yield` is the only coroutine keyword that accepts an expression and passes it to the promise. Dietmar Kühl's `yield_value` overload for `with_error<E> ([task.promise]/7[50])` is the best solution the language permits - and that is the problem. The constraints the sender model imposes are so narrow that even one of the most experienced C++ library engineers is left with a mechanism that reverses the established meaning of `co_yield` and requires a core language change to regularize. In every other coroutine context, `co_yield` means "produce a value and continue." Here it means "fail and terminate."

4.2 The Consequence

In regular C++, `co_return` delivers errors:

```

my_task< std::pair<std::error_code, std::string> >
read_body( tcp_socket& sock )
{
    char buf[1024];
    std::string body;
    for(;;)
    {
        auto [ec, n] = co_await sock.async_read_some(buf);
        body.append( buf, n );
        if( ec )
            co_return { ec, std::move(body) };
    }
}

```

P3552R3^[13] introduces a different mechanism:

```

std::execution::task<std::size_t>
do_read(tcp_socket& s, buffer& buf)
{
    auto [ec, n] = co_await s.async_read(buf);
    if (ec)
        co_yield with_error(ec);           // unfortunately, terminates the coroutine
    co_return n;
}

```

For six years, every C++ coroutine library has taught the same two conventions: `co_return` for final values, `co_yield` for intermediate values that the coroutine continues to produce. The second example breaks both: `co_yield` does not produce a value, and the coroutine does not continue.

4.3 Established Practice

No production C++ coroutine library uses `co_yield` for error signaling:

- cppcoro (Lewis Baker), Boost.Cobalt (Klemens Morgenstern), libcoro (Josh Baldwin): errors are exceptions or `co_return` values.
- folly::coro::Task (Facebook/Meta): errors are exceptions, propagated via `co_await`.
- Boost.Asio awaitable (Chris Kohlhoff): errors delivered as return values via `as_tuple(use_awaitable)`.

There is no third path. `std::execution::task` introduces one.

4.4 The Language Change

The `return_void` / `return_value` mutual exclusion has been part of the coroutine specification since N4499 (2015). In the decade since, cppcoro, folly::coro, Boost.Cobalt, Boost.Asio, libcoro, and asyncpp have shipped coroutine task types. None required the restriction to be lifted. All deliver errors through `co_return` or exceptions.

P1713R0^[59] (Baker, 2019) proposed removing this restriction. It had no consensus in Cologne.

P3950R0^[23] (Leahy, 2025) proposes the same change. The motivation is `std::execution`. The paper states: ***“Disallowing it either disadvantages coroutines vis-a-vis std::execution or necessitates library workarounds.”*** The paper targets EWG - a core language change - to serve a library adopted by LEWG.

Müller’s P3801R0^[19] (“Concerns about the design of `std::execution::task`,” 2025) confirms the cause: ***“The reason `co_yield` is used, is that a coroutine promise can only specify `return_void` or `return_value`, but not both. If we want to allow `co_return;`, we cannot have `co_return_with_error(error_code);`. This is unfortunate, but could be fixed by changing the language to drop that restriction.”***

A language rule that has served every coroutine library for a decade, and whose removal the committee previously declined, must now be reconsidered because one library requires it. The rule is not the problem.

5. Where Is the Allocator?

Coroutines dynamically allocate their frames. For I/O operations, the frame must outlive the launching function, so HALO cannot elide the allocation (Section 5.5). Niebler identified the cost^[33] in 2020:

“With coroutines, you have an allocation when a coroutine is first called, and an indirect function call each time it is resumed. The compiler can sometimes eliminate that overhead, but sometimes not.”

This is a real cost and a real source of resistance to coroutines in performance-sensitive code. But the allocator that matters here is not a general-purpose allocator. It is the **frame allocator** - an allocator whose sole purpose is to allocate and deallocate coroutine frames. Control and timing of the frame allocator is crucial: `promise_type::operator new` fires at the call site, before any sender machinery runs. With the right frame allocator, the overhead disappears. The resistance stems partly from the allocation model, not from coroutines themselves.

Each `task<T>` call invokes `promise_type::operator new`. At high request rates, frame allocations reach millions per second. A recycling frame allocator makes a measurable difference. P4003R0^[25] (“Coroutines for I/O”) benchmarks a 4-deep coroutine call chain (2 million iterations):

Platform	Frame Allocator	Time (ms)	Speedup
MSVC	Recycling	1265.2	3.1x
MSVC	mimalloc	1622.2	2.4x
MSVC	<code>std::allocator</code>	3926.9	-
Apple clang	Recycling	2297.08	1.6x
Apple clang	<code>std::allocator</code>	3565.49	-

In the sender model, `connect` / `start` binds the continuation after the frame is already allocated. The frame allocator arrives too late.

5.1 Senders

The P2300R10^[1] authors solved the allocator propagation problem for senders with care and precision. The receiver's environment carries the allocator via `get_allocator(get_env(rcvr))`, and the sender algorithms propagate it automatically through every level of nesting:

```
auto work =
    just(std::move(socket))
| let_value([](tcp_socket& s) {
    return async_read(s, buf)
    | let_value([&](auto data) {
        return parse(data)
        | let_value([&](auto doc) {
            return async_write(
                s, build_response(doc));
        });
    });
});

recycling_allocator<> alloc; // set once here
auto op = connect(
    write_env(std::move(work),
        prop(get_allocator, alloc)),
    rcvr);
start(op);
```

The frame allocator is set once at the launch site and reaches every operation automatically.

However...

Nothing here allocates. Used as intended, the operation state is a single concrete type with no heap allocation. The allocator propagates through a pipeline that does not need it.

5.2 Coroutines

A coroutine connection handler:

```
std::execution::task<>
handle_connection(tcp_socket conn)
{
    auto [ec, n] = co_await conn.async_read(buf);
    // ...
}
```


Spawned in the standard way:

```
ex::spawn(
    ex::on(sch, handle_connection(std::move(conn))), // operator new fires here
    scope.get_token());
```

`handle_connection(std::move(conn))` evaluates before `spawn` or `on`. No frame allocator can reach it.

The P2300 propagation mechanism is `write_env` with `get_allocator`:

```
recycling_allocator<> alloc;

auto op = connect(
    write_env(
        ex::on(sch,
            handle_connection(std::move(conn))), // operator new allocates here...
        prop(get_allocator, alloc)),           // ...but allocator arrives here
    rcvr);
start(op);
```

The receiver's environment carries the allocator after `connect`. The frame was allocated before.

The only way to get the frame allocator into the coroutine frame is through the parameter list:

```
template<class Allocator>
std::execution::task<>
handle_connection( std::allocator_arg_t, Allocator alloc, tcp_socket conn );
```

P2300's elegant environment propagation is structurally unreachable for coroutine frames.

Senders get the allocator they do not need. Coroutines need the frame allocator they do not get.

5.3 Coroutines Work for What Senders Get Free

While Kühl's P3552R3^[13] provides `std::allocator_arg_t` for the initial allocation, propagation remains unsolved. The child's `operator new` fires before the parent can intervene. The only workaround is manual forwarding. Three properties distinguish this from a minor inconvenience:

1. Composability loss. Generic sender algorithms like `let_value` and `when_all` launch child operations without knowledge of the caller's frame allocator. Manual forwarding cannot cross algorithm boundaries.
2. Silent fallback. Omitting `allocator_arg` from one call does not produce a compile error. The child silently falls back to the default heap allocator with no diagnostic.
3. Protocol asymmetry. Schedulers and stop tokens propagate automatically through the receiver environment. Frame allocators are the only execution resource that the sender model forces coroutine users to propagate by hand.

C++ Core Guidelines F.7^[37]: a function should not be coupled to the caller's ownership policy. Lampson (1983): *"An interface should capture the minimum essentials of an abstraction."*

Senders receive the frame allocator through the environment, automatically, at every level of nesting, with no signature pollution. Coroutines receive it through `allocator_arg`, manually, at every call site, with silent fallback on any mistake.

P3796R1^[18] Section 3.5.11 identifies the absence of a TLS capture/restore hook in `task`. P4003R0^[25] Section 5 demonstrates a working mechanism.

5.4 Does Performance Matter?

The recycling frame allocator eliminates coroutine frame allocation overhead. It requires `allocator_arg` at every call site. One missed site falls back to `std::allocator` with no diagnostic. In production code, sites will be missed. Users profile, see heap allocation cost, and conclude coroutines are slow. Coroutines are not slow. The fast path is too hard to use.

5.5 Will HALO Save Us?

HALO allows compilers to elide coroutine frame allocation when the frame's lifetime is provably bounded by its caller. When an I/O coroutine is launched onto an execution context, the frame must outlive the launching function:

```
namespace ex = std::execution;

ex::task<> read_data(socket& s, buffer& buf)
{
    co_await s.async_read(buf);
}

void start_read(ex::counting_scope& scope, auto sch, socket& sock, buffer& buf)
{
    ex::spawn(
        ex::on(sch, read_data(sock, buf)),
        scope.get_token());
}
```

The compiler cannot prove bounded lifetime, so HALO cannot apply and allocation is mandatory.

5.6 A Viral Signature?

Here is what frame allocator propagation looks like in a coroutine call chain under P3552R3^[13].

```

template<typename Allocator>
task<void> level_three(
    std::allocator_arg_t,
    Allocator alloc)
{
    co_return;
}

template<typename Allocator>
task<void> level_two(
    int x,
    std::allocator_arg_t,
    Allocator alloc)
{
    co_await level_three(std::allocator_arg, alloc);
}

template<typename Allocator>
task<int> level_one(
    int v,
    std::allocator_arg_t,
    Allocator alloc)
{
    co_await level_two(42, std::allocator_arg, alloc);
    co_return v;
}

```

5.7 Domain Freedom?

No workaround, global PMR, thread-local registries, or otherwise, can bypass the promise. Every allocated coroutine frame must run through `promise_type::operator new`. If the promise does not cooperate, the frame allocator does not reach the frame.

The escape hatch is to stop searching for a universal frame allocator model in one promise type. Let each domain's task type cooperate with its own frame allocator strategy. A networking task type can use thread-local propagation. A GPU task type can use device memory APIs. Solutions exist when the promise is free to serve its domain rather than forced to serve a model it does not participate in.

6. Where Is the Tail Call?

C++20 provides symmetric transfer (P0913R1^[62]): `await_suspend` returns a `coroutine_handle<>` and the compiler resumes the designated coroutine as a tail call. The stack does not grow. One coroutine suspends and another resumes in constant space. This is the mechanism C++20 provides to prevent stack overflow in coroutine chains. P2583R0^[63] ("Symmetric Transfer and Sender Composition") surveys six major production coroutine libraries. Five of six use symmetric transfer. The convergence is independent: different authors, different platforms, different design goals, the same mechanism.

P2300R10^[1]'s bridges between coroutines and the sender model use `void await_suspend`. When a coroutine co_awaits a sender, the `sender-awaiter` starts the sender inside `await_suspend`:

```
void await_suspend(coroutine_handle<Promise>) noexcept { start(state); }
```

When the sender completes synchronously, the receiver calls `.resume()` as a function call within `set_value`. The stack grows by one frame per synchronous completion. Sender algorithms (`then`, `let_value`, `when_all`) compose operations by wrapping one receiver in another. The wrapping receiver is a struct. It is not a coroutine. It has no `coroutine_handle<>`. No handle exists at any intermediate point in a sender pipeline. There is nothing to symmetric-transfer to.

Kühl documented this in P3796R1^[18], Section 3.2.3:

"The specification doesn't mention any use of symmetric transfer. Further, the `task` gets adapted by `affine_on` in `await_transform` (`[task.promise] p10`) which produces a different sender than `task` which needs special treatment to use symmetric transfer."

Kühl suggested domain customization of `affine_on` as a partial fix for the task-to-task case. Müller confirmed in P3801R0^[19], Section 3.1, that the partial fix does not reach the general case:

"while this would solve the example above, it would not solve the general problem of stack overflow when awaiting other senders. A thorough fix is non-trivial and requires support for guaranteed tail calls."

The only runtime mitigation is a trampoline scheduler that detects stack depth and reschedules. This is the queue-and-scheduler approach that P0913R1^[62] was specifically adopted to eliminate. P2583R0^[63] documents the full mechanism, surveys production practice, and argues the gap is architectural: sender algorithms are structs with void-returning completions. These are not boundary properties. They are what sender algorithms are. No launch mechanism in P3552R3^[13] avoids the sender composition layer - neither `sync_wait` nor `spawn(on(sch, task()), token)` - so the gap is inescapable from the entry point. The gap cannot be closed without removing the property it trades against - non-coroutine composition through sender algorithms - or by a language feature that does not exist. This is a tradeoff, not a defect. P2300R10^[1] was designed for GPU dispatch and heterogeneous computing - domains where symmetric transfer provides no benefit.

Symmetric transfer requires a handle. The sender model provides a function call.

7. The Gaps Are The Tradeoff

Niebler wrote in 2024: *"If your library exposes asynchrony, then returning a sender is a great choice: your users can await the sender in a coroutine if they like."*^[34] The phrase "if they like" implies this is straightforward. Niebler himself characterized^[33] the sender style as *"far harder to write and read than the equivalent coroutine."* The previous three sections suggest the cost is structural.

Each gap has the same shape. The sender model requires a property for compile-time analysis. That property forces a cost on coroutines at the boundary or inside the composition mechanism. The committee standardized the sender model for specific properties: compile-time routing, zero-allocation reification, type-level dispatch. Each gap documented in Sections 3-6 is the cost of one of those properties. Closing any gap requires removing the property it pays for.

7.1 The Channel Tradeoff

Section 3 showed the cost of compile-time channel routing. Choose one:

Compile-time channel routing	I/O tuple completion
<code>upon_error</code> , <code>let_error</code> , <code>upon_stopped</code> attach at the type level; algorithms dispatch on channel without runtime inspection	<code>(error_code, size_t)</code> returned together; error, cancellation, and byte count are all meaningful and inseparable

7.2 The `co_yield` Tradeoff

Section 4 showed the cost of a separate error channel for coroutines. The alternative, lifting the `return_void` / `return_value` mutual exclusion, is not part of C++26. Choose one:

Separate error channel from coroutines	Established <code>co_return</code> semantics
<code>co_yield with_error(ec);</code>	<code>co_return std::unexpected(ec);</code>

7.3 The Frame Allocator Tradeoff

Section 5 showed the cost of deferred execution. Choose one:

Deferred execution via <code>connect</code> / <code>start</code>	Frame allocator reaches coroutine frames
Receiver environment available after <code>connect()</code> ; zero-allocation sender pipelines	<code>promise_type::operator new</code> fires at the call site with the right frame allocator

`await_transform` cannot help: the child's `operator new` fires before `co_await` processing begins. P3552R3^[13] offers `allocator_arg` for the initial allocation, but propagation remains unsolved. P3826R3^[20] offers five solutions for algorithm dispatch; none changes when the frame allocator becomes available.

7.4 The Symmetric Transfer Tradeoff

Section 6 showed the cost of composing coroutines through sender algorithms. Choose one:

Coroutine composition through sender algorithms	Symmetric transfer
Sender algorithms are structs with void-returning completions; zero-allocation sender pipelines	<code>await_suspend</code> returns <code>coroutine_handle<></code> ; constant-stack coroutine chains

7.5 ABI Makes the Choice Permanent

The first four tradeoffs are structural. This subsection is about timing.

Once shipped, the three-channel model, the `connect / start` protocol, and the `void await_suspend` bridge become ABI. Every sender algorithm's behavior is defined in terms of which channel fires. The relationship between the promise's `operator new` and `connect()` becomes fixed. The void-returning `await_suspend` in `sender-awaitable` becomes fixed. Closing any of the four gaps after standardization requires changing these relationships - a breaking change to the sender protocol.

Choose one:

Ship <code>task</code> in C++26	Iterate the coroutine integration
The four tradeoffs above become ABI	The four tradeoffs remain open

The natural compromise, ship `task` with known limitations and fix via DR or C++29 addendum, assumes the fix is a minor adjustment. Sections 7.1 through 7.4 show it is not. Each gap is the cost of a specific design property. Closing the gap means removing the property.

Shipping `task` is the risky choice, not the safe one. Four structural gaps locked in by ABI. A language change proposed to fix `co_yield with_error` for a channel I/O never uses. Frame allocator propagation still unsolved six months after adoption. Symmetric transfer structurally unreachable through the sender pipeline. Kühl himself filed a catalogue of open concerns (Section 9). The sender model's environment leaks into the coroutine's public type: P3552R3^[13] defines `task` with two template parameters where every production coroutine library has independently converged on one:

Library	Declaration	Params
asyncpp	<code>template<class T> class task</code>	1
Boost.Cobalt	<code>template<class T> class task</code>	1
cppcoro	<code>template<typename T> class task</code>	1
aiopp	<code>template<typename Result> class Task</code>	1
libcoro	<code>template<typename return_type> class task</code>	1
P3552R3 (std)	<code>template<class T, class Environment> class task</code>	2

Sources: [cppcoro](#)^[42], [libcoro](#)^[47], [asyncpp](#)^[46], [aiopp](#)^[48], [P3552R3 ref. impl.](#)^[49]

Deferring `task` to C++29 costs nothing: no production user depends on it, every networking library already ships its own task type, and C++29 forwarding was unanimous.

The narrowest remedy is to ship `std::execution` without `task`. The sender pipeline is valuable and ready. The four gaps exist at the coroutine boundary and inside the composition mechanism. Remove `task` from C++26, and no production user is affected. The cost falls on no one.

8. Why Wait To Ship?

The cost of shipping is documented above. What is the cost of waiting? The cost is real. C++26 ships without a standard task, and the committee's work on coroutine integration does not reach users in this cycle. The question is whether that cost exceeds the cost of locking in the four gaps documented above.

8.1 "C++ needs a standard task. Six years is long enough."

C++ needs `std::task`. The question is whether `std::execution::task` is that type.

Convention	P3552R3	Every existing library
Error signaling	<code>co_yield with_error(ec)</code>	<code>co_return</code> or exceptions
Type signature	<code>task<T, Environment></code>	<code>task<T></code>

A task bound to one model's conventions is a task for that model. The standard task for C++ has not been written yet.

8.2 "The gaps are manageable. Ship now, iterate later."

"Fix later" assumes the fix is a minor adjustment.

Each gap is the cost of a specific design property (Section 7). Closing the gap means removing the property. ABI lock-in makes the choice permanent (Section 7.5). The committee deferred P2300 from C++23 for the same pattern of ongoing design changes.

The committee has been here before.

8.3 "A standard task enables library interop that no third-party type can."

Coroutine interop requires the awaitable protocol, not type identity.

P3552R3^[13] Section 3:

- "different coroutine task implementations can live side by side: not all functionality has to be implemented by the same coroutine task."
- "it should be possible to `co_await` awaitables which includes both library provided and user provided ones."

Open task types ship in `cppcoro`^[42], `Boost.Cobalt`^[41], `libunifex`^[43], `folly::coro`^[44], `QCoro`^[45], and `asyncpp`^[46]. Each interoperates through the awaitable protocol.

8.4 "Without `task`, coroutine users cannot access standard networking."

No standard networking exists in C++26. Every networking library that supports coroutines already ships its own task type:

Library	Coroutine Type
<code>Boost.Asio</code>	<code>awaitable<T></code>

Library	Coroutine Type
Boost.Cobalt	task , promise
folly::coro	Task<T>
libcoro	task<T>
COROIO	TTask
asyncpp	task

Coroutine users are not locked out of networking today and will not be locked out tomorrow.

Each argument assumes `std::execution::task` is the standard task C++ needs. The evidence suggests it is the execution model's task - and that the standard task for C++ is still ahead.

9. Maturity of the Coroutine Integration

The coroutine integration is not ready for C++26. Three data points:

LEWG polled the frame allocator question directly (P3796R1^[18], September 2025):

"We would like to use the allocator provided by the receivers env instead of the one from the coroutine frame"

SF	F	N	A	SA
0	0	5	0	0

Attendance: 14. Outcome: strictly neutral.

The entire room abstained. Without a mechanism to propagate frame allocator context through nested coroutine calls, the committee had no direction to endorse. Kühl himself identified the need for further iteration, reworking the frame allocator propagation model in D3980R0^[24] (2026-01-25), six months after P3552R3^[13]'s adoption at Sofia. LWG 4356 confirms the gap has been filed as a specification defect.

The task type's forwarding polls (LEWG, 2025-05-06):

"Forward P3552R1 to LWG for C++29"

SF:5 / F:7 / N:0 / A:0 / SA:0 - unanimous.

"Forward P3552R1 to LWG with a recommendation to apply for C++26 (if possible)."

SF:5 / F:3 / N:4 / A:1 / SA:0 - weak consensus, with "if possible" qualifier.

C++29 forwarding was unanimous. C++26 was conditional and weak. With the thoroughness that has defined his stewardship of `task`, Kühl cataloged sixteen open concerns in P3796R1^[18] ("Coroutine Task Issues").

10. Conclusion

This paper recommends: ship `std::execution` for C++26, defer `task` to C++29, and explore coroutine-native I/O designs alongside sender-based designs.

`std::execution` has earned its place. Herb Sutter reported that Citadel Securities uses it in production: *"We already use C++26's `std::execution` in production for an entire asset class, and as the foundation of our new messaging infrastructure."*^[36] Senders work well in their domain: compile-time work graph construction, GPU dispatch, high-frequency trading pipelines. Even GPU cloud APIs are accessed through HTTP. The finding is not that `std::execution` failed. It is that its scope is specific, not universal. Narrowing the scope is not admitting failure - it is recognizing success where it exists and clearing the path where it does not.

The four gaps documented in Sections 3-6 are the cost of treating the sender model as the universal model of asynchronous computation. Each gap is the cost of a property the sender model requires. They are not design defects. They are tradeoffs - and coroutine I/O should not be forced to pay them.

1. Is the sender model a universal model of asynchronous computation, or a domain-specific one? The evidence suggests it serves GPU dispatch, high-frequency trading, and heterogeneous computing, but not I/O.
2. Are coroutines the primary tool for asynchronous C++? If yes, the coroutine integration deserves the same design investment as the sender pipeline itself.
3. Should `task<T>` ship in C++26 with these costs? Ship `std::execution` for the domains it serves. Let the coroutine integration iterate independently.

The SG4 poll (Kona 2023, SF:5/F:5/N:1/A:0/SA:1) presented two alternatives. A coroutine-native approach was not among the choices. The straw polls in Section 11 ask the committee whether the choice should remain binary.

11. Suggested Straw Polls

1. "I/O completions that carry both an error code and a byte count present a design challenge for the three-channel completion model."
 2. "The coroutine integration in `std::execution` has open design questions that would benefit from further iteration."
 3. "WG21 should explore coroutine-native I/O designs alongside sender-based designs."
-

Appendix A - The Three-Channel Model

A.1 Why the Three-Channel Model Exists

The sender model constructs the entire work graph at compile time as a deeply-nested template type. `connect(sndr, rcvr)`^[50] collapses the pipeline into a single concrete type. For this to work, every control flow path must be distinguishable at the type level, not the value level.

The three completion channels provide exactly this. **Completion signatures**^[50] declare three distinct type-level paths:

```
using completion_signatures =
    std::execution::completion_signatures<
        set_value_t(int),                // value path
        set_error_t(std::error_code),    // error path
        set_stopped_t()>;              // stopped path
```

Algorithms dispatch on which channel fired without inspecting payloads. `upon_error` attaches to the error path at the type level. `let_value` attaches to the value path at the type level. `upon_stopped` attaches to the stopped path. The routing is in the types, not in the values.

If errors were delivered as values (for example, `expected<int, error_code>` through `set_value`), the compiler would see one path carrying one type. Algorithms could not dispatch on error without runtime inspection of the payload. Every algorithm would need runtime branching logic to inspect the expected and route accordingly.

The three channels exist because the sender model is a compile-time language.

Compile-time languages route on types. Runtime languages route on values. A coroutine returns

`auto [ec, n] = co_await read(buf)` and branches with `if (ec)` at runtime. The sender model encodes `set_value` and `set_error` as separate types in the completion signature and routes at compile time. The three-channel model is not an arbitrary design choice. It is a structural requirement of the compile-time work graph.

A compile-time language cannot express partial success. I/O operations return `(error_code, size_t)` together because partial success is normal. The three-channel model demands that the sender author choose one channel. No choice is correct because the compile-time type system cannot represent “both at once.”

Eliminating the three-channel model would remove the type-level routing that makes the compile-time work graph possible. The three channels are not a design flaw. They are the price of compile-time analysis. I/O cannot pay that price because I/O’s completion semantics are inherently runtime.

A.2 `async_read` as a Sender

Section 3.3 shows that published sender code loses partial results. Here is the complete sender implementation using

P2300R10 Section 1.4^[1]’s `recv_sender` and `async_recv` :

```

template<class Rcvr>
struct read_op
{
    struct rcvr_rcvr {
        read_op* self_;
        void set_value(std::size_t n) && noexcept { self_->completed({}, n); }
        void set_error(std::error_code ec) && noexcept { self_->completed(ec, 0); }
        void set_stopped() && noexcept { execution::set_stopped(std::move(self_->rcvr_)); }
    };

    Rcvr rcvr_;
    SOCKET sock_;
    std::byte* data_;
    std::size_t len_;
    std::size_t bytes_read_ = 0;
    std::optional<connect_result_t<rcvr_sender, rcvr_rcvr>> child_;

    void start() & noexcept { start_rcvr(); }

    void start_rcvr() {
        child_.emplace(execution::connect(
            async_rcv(sock_, data_ + bytes_read_, len_ - bytes_read_),
            rcvr_rcvr{this}));
        execution::start(*child_);
    }

    void completed(std::error_code ec, std::size_t n) {
        bytes_read_ += n;
        if (bytes_read_ == len_)
            return execution::set_value(std::move(rcvr_), bytes_read_);
        if (!ec)
            return start_rcvr();
        execution::set_error(std::move(rcvr_), ec); // bytes_read_ bytes are lost
    }
};

```

Every name is from P2300R10: `rcvr_sender`, `async_rcv`, `connect`, `start`, `set_value`, `set_error`, `set_stopped`, `connect_result_t`. The `rcvr_rcvr` is the standard pattern for wiring a child sender back to a parent operation state.

A.3 Timeline of the Error Channel Question

The partial success problem was raised independently, across multiple groups, by participants with different domain backgrounds, over a span of five years. Every proposed resolution carried real costs. The question was never resolved - not for lack of attention, but because the three-channel model may not admit a solution that preserves both compile-time routing and I/O's tuple semantics.

- 2020 (Nov): Niebler's "[Structured Concurrency](#)"^[33] blog post establishes the three-channel model publicly. Acknowledges CPS is harder to write and read than coroutines.

- 2020 (Feb, Prague): During review of P1678R2^[56] (“Callbacks and Composition”), a participant raised that asynchronous operations need not fail completely or succeed completely, and that no pattern in the proposal supports partial success. The response was that nothing would be different. LEWG polled to encourage more work (SF:7/F:14/N:9/A:3/SA:0). No resolution on partial success emerged.
- 2021 (Feb, SG4 telecon): During review of P0958R3, a participant stated that sender/receivers have a loss because they do not have success/partial-success. Another suggested adapting from success/error; the response was that the error does not carry information. No poll on the error channel design.
- 2021 (Jul-Oct, LEWG telecon series): The partial success question was debated across at least five LEWG telecons during the P2300 review. Multiple participants raised incompatible positions. One argued that partial success calling `set_value` does not work for generic retry algorithms. Others described `set_error` as effectively `set_exception` - an error channel that does not serve error conditions. The August 17, 2021 LEWG outcome document explicitly listed “Better explain how partial success works with senders/receivers” as open guidance to the P2300 authors - an acknowledgment that the question was unresolved. During the October 4, 2021 LEWG telecon, a participant (who gave the authors permission to anonymously quote what they said) indicated that partial success could be addressed through async streams once the initial sender/receiver facilities were in place, with the expectation that such streams could then be standardized. The async streams facility was never proposed, never standardized, and is not in the C++26 working draft.
- 2021 (Aug): Kohlhoff published P2430R0^[3] (“Partial success scenarios with P2300”), slides from a presentation during the P2300 review, identifying that partial results must be communicated down the `set_value` channel due to limitations of `set_error` and `set_done`. Independent confirmation of the same tension the preceding telecons debated, from the author of Asio.
- 2022-2024: P2300R4^[2] through P2300R10^[1] published. Three-channel model unchanged across all revisions.
- 2023: P2762R2^[4] (Dietmar Kühl, “Sender/Receiver Interface for Networking”) preserves the proven `(error_code, size_t)` convention from Asio - implicitly acknowledging I/O needs both values together.
- 2023 (Nov, Kona): SG4 polls that networking must use the sender model (SF:5/F:5/N:1/A:0/SA:1). Every future I/O operation must now navigate the channel mismatch.
- 2024 (Nov, Wroclaw): The same three-channel question resurfaced during design of P0260 (“C++ Concurrent Queues”). LEWG debated whether a closed queue should complete through `set_error`, `set_value`, or `set_stopped`. One participant asked how popping from a closed queue could be an error when it represents the ultimate success of processing all items. Another noted that POSIX does not model EOF this way - reading from a socket at EOF succeeds with zero bytes. The debate consumed portions of two face-to-face meetings.
- 2025 (Feb, Hagenberg): The concurrent queue channel question remained open. A poll to reopen the channel choice was withdrawn. A new problem was discovered: `try_push` cannot safely schedule an `async_pop` continuation because `std::execution` provides no way to express that a schedule operation is non-blocking - a concrete manifestation of the architectural mismatch in a producer-consumer API.
- 2025-2026: P2300 adopted into the C++26 working draft. P3570R2^[15] (Fracassi, “Optional variants in sender/receiver”) documents the interface mismatch between `optional<T>` and the channel model. The error channel / partial success question remains open. No paper in the published record proposes a resolution that preserves compile-time channel routing.

The question has been open for five years. This duration is not due to neglect - some of the most capable engineers in the C++ community have worked on P2300 across ten revisions and dozens of follow-on papers. Every proposed fix involves a tradeoff with real downsides (Sections 3.3 through 3.8 document the costs). The three-channel model is a property of the sender model, not a bug. When the best people in the field iterate for five years on a well-understood problem and every solution trades one cost for another, the evidence points toward a structural constraint rather than a missing insight.

Appendix B - The Frame Allocator Gap

B.1 The Full Ceremony for Frame-Allocator-Aware Coroutines

The sender model requires five layers of machinery to propagate a custom frame allocator through a coroutine call chain:

```

namespace ex = std::execution;

// 1. Define a custom environment that answers get_allocator
struct my_env
{
    using allocator_type = recycling_allocator<>;
    allocator_type alloc;

    allocator_type query(ex::get_allocator_t) const noexcept
    {
        return alloc;
    }
};

// 2. Alias the task type with the custom environment
using my_task = ex::task<void, my_env>;

// 3. Every coroutine accepts and forwards the frame allocator
template<typename Allocator>
my_task level_two(
    int x,
    std::allocator_arg_t,
    Allocator alloc)
{
    co_return;
}

template<typename Allocator>
my_task level_one(
    int v,
    std::allocator_arg_t,
    Allocator alloc)
{
    co_await level_two(42, std::allocator_arg, alloc);
    co_return;
}

// 4. Construct the environment at the launch site
// 5. Inject it via write_env
void launch(ex::counting_scope& scope, auto sch)
{
    recycling_allocator<> alloc; // step 4
    my_env env{alloc};
    auto sndr =
        ex::write_env(
            level_one(0, std::allocator_arg, alloc), env) // step 5
        | ex::continues_on(sch);
    ex::spawn(std::move(sndr), scope.get_token());
}

```

Forgetting any one of the five steps silently falls back to the default allocator. The compiler provides no diagnostic.

Appendix C - Direction of Change

The claim is not that the volume of changes is abnormal; it is that the direction is uniform. Every paper, LWG issue, and NB comment modifying `std::execution` since Tokyo (March 2024) falls into one of two categories.

Sender Sub-Language items address the CPS model's own machinery: algorithm customization, operation state lifetimes, completion signature constraints, removals of primitives that violated structured concurrency.

Sender Integration items address the boundary where the CPS model meets coroutines: the `task` type, frame allocator propagation into coroutine frames, `co_yield with_error` semantics.

Coroutine-Intrinsic items are specific to coroutines. There are none.

Origin	Items
Sender Sub-Language	P2855R1, P2999R3, P3175R3, P3187R1, P3303R1, P3373R2, P3557R3, P3570R2, P3682R0, P3718R0, P3826R3, P3941R1, LWG 4190, LWG 4206, LWG 4215, LWG 4368
Sender Integration	P3927R0, P3950R0, D3980R0, LWG 4356, US 255-384, US 253-386, US 254-385, US 261-391
Coroutine-Intrinsic	-

All data is gathered from the published [WG21 paper mailings](#)^[54], the [LWG issues list](#)^[55], and the [C++26 national body ballot comments](#)^[53].

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References

WG21 Papers

1. [P2300R10](https://wg21.link/p2300r10) - “std::execution” (Michał Dominiak, Georgy Evtushenko, Lewis Baker, Lucian Radu Teodorescu, Lee Howes, Kirk Shoop, Michael Garland, Eric Niebler, Bryce Adelstein Lelbach, 2024). <https://wg21.link/p2300r10>
2. [P2300R4](https://wg21.link/p2300r4) - “std::execution” (Michał Dominiak, et al., 2022). <https://wg21.link/p2300r4>
3. [P2430R0](https://wg21.link/p2430r0) - “Partial success scenarios with P2300” (Chris Kohlhoff, 2021). <https://wg21.link/p2430r0>
4. [P2762R2](https://wg21.link/p2762r2) - “Sender/Receiver Interface For Networking” (Dietmar Kühl, 2023). <https://wg21.link/p2762r2>
5. [P2855R1](https://wg21.link/p2855r1) - “Member customization points for Senders and Receivers” (Ville Voutilainen, 2024). <https://wg21.link/p2855r1>
6. [P2999R3](https://wg21.link/p2999r3) - “Sender Algorithm Customization” (Eric Niebler, 2024). <https://wg21.link/p2999r3>
7. [P3149R11](https://wg21.link/p3149r11) - “async_scope” (Ian Petersen, Jessica Wong, Kirk Shoop, et al., 2025). <https://wg21.link/p3149r11>
8. [P3164R4](https://wg21.link/p3164r4) - “Improving Diagnostics for Sender Expressions” (Eric Niebler, 2024). <https://wg21.link/p3164r4>
9. [P3175R3](https://wg21.link/p3175r3) - “Reconsidering the std::execution::on algorithm” (Eric Niebler, 2024). <https://wg21.link/p3175r3>
10. [P3187R1](https://wg21.link/p3187r1) - “Remove ensure_started and start_detached from P2300” (Kirk Shoop, Lewis Baker, 2024). <https://wg21.link/p3187r1>
11. [P3303R1](https://wg21.link/p3303r1) - “Fixing Lazy Sender Algorithm Customization” (Eric Niebler, 2024). <https://wg21.link/p3303r1>
12. [P3373R2](https://wg21.link/p3373r2) - “Of Operation States and Their Lifetimes” (Robert Leahy, 2025). <https://wg21.link/p3373r2>
13. [P3552R3](https://wg21.link/p3552r3) - “Add a Coroutine Task Type” (Dietmar Kühl, Maikel Nadolski, 2025). <https://wg21.link/p3552r3>
14. [P3557R3](https://wg21.link/p3557r3) - “High-Quality Sender Diagnostics with Constexpr Exceptions” (Eric Niebler, 2025). <https://wg21.link/p3557r3>
15. [P3570R2](https://wg21.link/p3570r2) - “Optional variants in sender/receiver” (Fabio Fracassi, 2025). <https://wg21.link/p3570r2>
16. [P3682R0](https://wg21.link/p3682r0) - “Remove std::execution::split” (Robert Leahy, 2025). <https://wg21.link/p3682r0>
17. [P3718R0](https://wg21.link/p3718r0) - “Fixing Lazy Sender Algorithm Customization, Again” (Eric Niebler, 2025). <https://wg21.link/p3718r0>
18. [P3796R1](https://wg21.link/p3796r1) - “Coroutine Task Issues” (Dietmar Kühl, 2025). <https://wg21.link/p3796r1>
19. [P3801R0](https://wg21.link/p3801r0) - “Concerns about the design of std::execution::task” (Jonathan Müller, 2025). <https://wg21.link/p3801r0>
20. [P3826R3](https://wg21.link/p3826r3) - “Fix Sender Algorithm Customization” (Eric Niebler, 2026). <https://wg21.link/p3826r3>
21. [P3927R0](https://wg21.link/p3927r0) - “task_scheduler Support for Parallel Bulk Execution” (Lee Howes, 2026). <https://wg21.link/p3927r0>
22. [P3941R1](https://wg21.link/p3941r1) - “Scheduler Affinity” (Dietmar Kühl, 2026). <https://wg21.link/p3941r1>
23. [P3950R0](https://wg21.link/p3950r0) - “return_value & return_void Are Not Mutually Exclusive” (Robert Leahy, 2025). <https://wg21.link/p3950r0>
24. [D3980R0](https://isocpp.org/files/papers/D3980R0.html) - “Task’s Allocator Use” (Dietmar Kühl, 2026). <https://isocpp.org/files/papers/D3980R0.html>
25. [P4003R0](https://wg21.link/p4003r0) - “Coroutines for I/O” (Vinnie Falco, Steve Gerbino, Mungo Gill, 2026). <https://wg21.link/p4003r0>
26. [P4014R0](https://wg21.link/p4014r0) - “The Sender Sub-Language” (Vinnie Falco, 2026). <https://wg21.link/p4014r0>
27. [N5028](https://wg21.link/n5028) - “Result of voting on ISO/IEC CD 14882” (2025). <https://wg21.link/n5028>

LWG Issues

28. [LWG 4368](https://cplusplus.github.io/LWG/issue4368) - “Potential dangling reference from `transform_sender`” (Priority 1). <https://cplusplus.github.io/LWG/issue4368>
29. [LWG 4206](https://cplusplus.github.io/LWG/issue4206) - “`connect_result_t` should be constrained with `sender_to`” (Priority 1). <https://cplusplus.github.io/LWG/issue4206>

- 30. **LWG 4190** - “ `completion-signatures-for` specification is recursive” (Priority 2).
<https://cplusplus.github.io/LWG/issue4190>
- 31. **LWG 4215** - “ `run_loop::finish` should be `noexcept` ”. <https://cplusplus.github.io/LWG/issue4215>
- 32. **LWG 4356** - “ `connect()` should use `get_allocator(get_env(rcvr))` ”. <https://cplusplus.github.io/LWG/issue4356>

Blog Posts

- 33. Eric Niebler, “**Structured Concurrency**” (2020). <https://ericniebler.com/2020/11/08/structured-concurrency/>
- 34. Eric Niebler, “**What are Senders Good For, Anyway?**” (2024). <https://ericniebler.com/2024/02/04/what-are-senders-good-for-anyway/>
- 35. Herb Sutter, “**Trip report: Summer ISO C++ standards meeting (St Louis, MO, USA)**” (2024).
<https://herbsutter.com/2024/07/02/trip-report-summer-iso-c-standards-meeting-st-louis-mo-usa>
- 36. Herb Sutter, “**Living in the future: Using C++26 at work**” (2025). <https://herbsutter.com/2025/04/23/living-in-the-future-using-c26-at-work>
- 37. **C++ Core Guidelines** - F.7, R.30 (Bjarne Stroustrup, Herb Sutter, eds.).
<https://isocpp.github.io/CppCoreGuidelines/CppCoreGuidelines.html>
- 38. Butler Lampson, “**Hints for Computer System Design**” (1983). <https://bwlampson.site/33-Hints/Acrobat.pdf>
- 39. **Continuation-passing style** - Wikipedia. https://en.wikipedia.org/wiki/Continuation-passing_style

Libraries

- 40. **Boost.Asio** - Asynchronous I/O library (Chris Kohlhoff). https://www.boost.org/doc/libs/release/doc/html/boost_asio.html
- 41. **Boost.Cobalt** - Coroutine task types for Boost (Klemens Morgenstern).
<https://www.boost.org/doc/libs/develop/libs/cobalt/doc/html/index.html>
- 42. **cppcoro** - A library of C++ coroutine abstractions (Lewis Baker). <https://github.com/lewissbaker/cppcoro>
- 43. **libunifex** - Unified Executors library for C++ (Facebook/Meta, Eric Niebler).
<https://github.com/facebookexperimental/libunifex>
- 44. **folly::coro** - Facebook’s coroutine library. <https://github.com/facebook/folly/tree/main/folly/experimental/coro>
- 45. **QCoro** - Coroutine integration for Qt (Daniel Vrátil). <https://qcoro.dev/>
- 46. **asyncpp** - Async coroutine library (Péter Kardos). <https://github.com/petiaccja/asyncpp>
- 47. **libcoro** - C++20 coroutine library (Josh Baldwin). <https://github.com/jbaldwin/libcoro>
- 48. **aiopp** - Async I/O library (Joel Schumacher). <https://github.com/pfirsich/aiopp>
- 49. **beman::task** - P3552R3 reference implementation (Beman Project). <https://github.com/bemanproject/task>

Other

- 50. **C++ Working Draft** - (Richard Smith, ed.). <https://eel.is/c++draft/>
- 51. **cppreference** - C++ reference documentation. <https://en.cppreference.com/>
- 52. **stdexec** - NVIDIA reference implementation of std::execution. <https://github.com/NVIDIA/stdexec>

- 53. **C++26 NB ballot comments** - National body comments repository. <https://github.com/cplusplus/nbballot>
- 54. **WG21 paper mailings** - ISO C++ committee papers. <https://open-std.org/jtc1/sc22/wg21/docs/papers/>
- 55. **LWG issues list** - Library Working Group active issues. <https://cplusplus.github.io/LWG/lwg-toc.html>
- 56. **P1678R2** - “Callbacks and Composition” (Kirk Shoop, 2020). <https://wg21.link/p1678r2>
- 57. **cplusplus/sender-receiver #247** - “Add ability to know when computing completion-signatures whether the call to connect() will throw” (Lewis Baker, 2024). <https://github.com/cplusplus/sender-receiver/issues/247>
- 58. **P3388R2** - “Provide nothrow connect guarantee for P2300” (Lewis Baker, 2025). <https://wg21.link/p3388r2>
- 59. **P1713R0** - “Allowing both co_return; and co_return value; in the same coroutine” (Lewis Baker, 2019). <https://wg21.link/p1713r0>
- 60. Robert Leahy, **use_sender.hpp** - Asio-to-sender bridge in stdexec (2026). https://github.com/NVIDIA/stdexec/blob/9d5836a634a21ecb06d17352905d04f99f635be6/include/asioexec/use_sender.hpp
- 61. **“Design Rationale: Sender Channels and I/O Return Types”** - Channel routing alternatives for LEWG (2026). <https://github.com/cppalliance/capy/blob/eb1de34a93b64b49e2c8826ca5088bdf72e1e1eb/doc/sender-channels-rationale.md>
- 62. **P0913R1** - “Add symmetric coroutine control transfer” (Gor Nishanov, 2018). <https://wg21.link/p0913r1>
- 63. **P2583R0** - “Symmetric Transfer and Sender Composition” (Mungo Gill, Vinnie Falco, 2026). <https://wg21.link/p2583r0>